

Task 1 Pipeline Overview

PixelPulse addresses GAVE2 Task 1 with a CFP-only topology-oriented ensemble. Five fold-specific HRVRL PGNet models provide the principal artery, vessel, and vein probability maps. A public RRWNet checkpoint supplies one refinement stage per fold. The two model families are averaged, blended with fixed weights, and passed through deterministic component filtering, morphological closing, and skeleton-guided gap repair. The final preliminary Task 1 score was 8.16666.

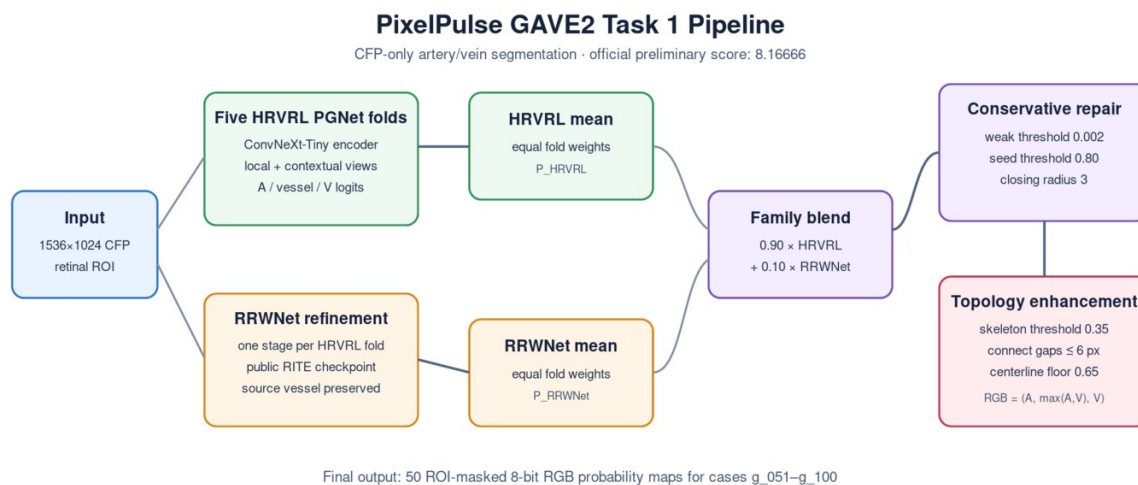


Fig. 1. PixelPulse Task 1 pipeline. $P(i)$ denotes the prediction of HRVRL fold i , and $R(i)$ denotes its one-stage RRWNet refinement.

1. Task 1: CFP-Only Artery/Vein Segmentation

1.1 Dataset and Data Division

The official preliminary dataset contains 50 labelled training cases (g_001 to g_050) and 50 unlabelled preliminary cases (g_051 to g_100). Each case contains a 1536 by 1024 colour fundus photograph and a retinal region-of-interest mask. Task 1 uses the CFP and ROI only. FFA images are not loaded during training, validation, inference, refinement, or post-processing.

Table 1. Case-level five-fold data division used for Task 1.

Subset	Cases	Use
Training pool	50, g_001 to g_050	Five-fold model development
Per-fold training	40	Parameter optimisation
Per-fold validation	10	Checkpoint selection
Preliminary set	50, g_051 to g_100	Leaderboard inference only

The split is deterministic and case-level. Each labelled case appears in one held-out fold, so patches from one image cannot occur in both training and validation within a fold. The split manifest SHA-256 is a182eac3715521d7b4094b5fd368b853341e330419319b37e9da0d0f155740. The selected checkpoint maximises the weighted criterion below.

$$S_{sel} = 0.50 DSC_A + 0.15 DSC_{BV} + 0.35 DSC_V \quad (1)$$

1.2 Input, Pre-processing, and Augmentation

Training uses a 512 by 512 local CFP crop and an aligned 768 by 768 contextual crop resized to 512 by 512. The contextual view provides vascular geometry, while the local view retains thin-vessel detail. Eight crops are sampled per case per epoch. A crop is vessel-centred with probability 0.80 and otherwise centred at a random ROI pixel. Images use reflection padding; labels, ROI masks, and teacher maps use zero padding.

CFP intensities are scaled to [0, 1] and normalised channel-wise using ImageNet mean $\mu = (0.485, 0.456, 0.406)$ and standard deviation $\sigma = (0.229, 0.224, 0.225)$.

$$x'_c = (x_c - \mu_c) / \sigma_c \quad (2)$$

The local crop, contextual crop, label, ROI, and teacher target receive the same random horizontal flip, vertical flip, and rotation from {0, 90, 180, 270} degrees. CFP crops also receive shared contrast sampled from [0.92, 1.08] and brightness sampled from [-0.04, 0.04]. No CLAHE, histogram matching, colour transfer, or FFA registration is used.

1.3 Network Architecture and Objective

The principal model is HRVRL PGNet with a ConvNeXt-Tiny encoder, paired local and contextual inputs, local-global semantic fusion, and centerness enabled. The network has 28,923,950 parameters and produces three logits ordered as artery, vessel union, and vein. The public G_pretrain.pkl checkpoint is loaded with complete compatible-key coverage.

The supervised objective combines ROI-masked weighted binary cross-entropy, soft Dice, and focal Tversky losses. Channel weights are (1.25, 0.75, 1.00) for artery, vessel, and vein. Focal Tversky uses alpha 0.35, beta 0.65, and gamma 0.75. A fixed zero-shot HRVRL teacher contributes weighted MSE, and a hierarchy term penalises artery or vein probability above vessel probability.

$$L_{total} = 0.30 L_{BCE} + 0.55 L_{Dice} + 0.15 L_{FT} + 0.10 L_{teacher} + 0.02 L_{hierarchy} \quad (3)$$

1.4 Training Protocol

Table 2. Training and inference settings for the selected five-fold HRVRL models.

Item	Setting	Item	Setting
Maximum epochs	24	Batch size	2
Patches per case	8	Gradient accumulation	4
Optimiser	AdamW	Weight decay	1e-4
Decoder learning rate	1e-4	Encoder learning rate	1e-5
Encoder freeze	5 epochs	Unfrozen blocks	Final 2
Validation interval	2 epochs	Validation threshold	0.45
Inference window	512	Inference stride	150
Mixed precision	Enabled	Random seed	20260723

Fold checkpoints were selected at epochs 13, 15, 21, 15, and 21. Their held-out mean artery, vein, and vessel Dice values were 0.7459, 0.7932, and 0.7877, respectively. Mixed precision and gradient-norm clipping at 1.0 were enabled. Learning rates were reduced on validation plateaus.

1.5 Inference, Refinement, and Ensemble

Each fold predicts every preliminary image by overlapping 512 by 512 windows with stride 150. Fold outputs are averaged equally. Each fold map is also converted from GAVE2 order (artery, vessel, vein) to RRWNet order (artery, vein, vessel), refined once, converted back, and averaged. The original vessel channel is preserved inside the RRWNet adapter. The family blend is fixed at 90 percent HRVRL and 10 percent RRWNet.

$$P_{raw} = 0.90 (1/5 \sum_{i=1}^5 P_i) + 0.10 (1/5 \sum_{i=1}^5 R_i) \quad (4)$$

1.6 Post-processing

Post-processing is deterministic and has two stages. First, weak support is defined at probability 0.002 and strong seeds at 0.80. Eight-connected weak components without a strong seed are removed, and binary closing with radius 3 repairs short discontinuities. Artery and vein remain non-exclusive.

Second, artery and vein support at threshold 0.35 is skeletonised. Skeleton endpoints separated by 1.5 to 6.0 pixels are connected by an ROI-constrained line, and existing and connected centerlines are raised to probability 0.65. The final vessel channel is reconstructed as $\max(\text{artery}, \text{vein})$, then all channels are quantised to an ROI-masked 8-bit RGB PNG.

2. Experimental Results

Table 3. Official preliminary Task 1 result and organizer baseline.

Method	Class	AV DSC	Sen	Spe	Acc	INF	COR
Baseline	Artery	0.7656	0.6709	0.9937	0.9828	0.6924	0.3058
Baseline	Vein	0.7656	0.7146	0.9930	0.9808	0.7672	0.2322
PixelPulse	Artery	0.6702	0.9521	0.9589	0.9587	0.2614	0.7290
PixelPulse	Vein	0.6702	0.9639	0.9569	0.9572	0.2290	0.7558

PixelPulse obtained an official Task 1 score of 8.16666, an absolute gain of 1.96276 over the published baseline score of 6.20390. The method exchanges some Dice and specificity for much higher sensitivity and substantially better path topology. This is favourable because classification and topology contribute 80 percent of the official Task 1 score.

$$S_{T1} = 2 D_{AV} + (4/6) \Sigma_{c \in \{A,V\}} (Sen_c + Spe_c + Acc_c) + \Sigma_{c \in \{A,V\}} (1 - INF_c + COR_c) \quad (5)$$

Using the displayed metrics, the Dice, classification, and topology contributions are 1.3404, 3.8318, and 2.9944. Their sum is 8.1666; the organizer reports the unrounded value 8.16666.

Table 4. Official-server ablation of the selected Task 1 branch.

Candidate	AV DSC	Task 1 score	Change
HRVRL plus 10% RRWNet plus closing	0.6264	8.07853	Reference
Add skeleton topology enhancement	0.6702	8.14713	+0.06860
Same enhanced payload, repeat evaluation	0.6702	8.16666	+0.08813

The same enhanced Task 1 bytes received 8.14713 and 8.16666 in repeated evaluations. Dice and pixel metrics were effectively unchanged, while INF and COR varied slightly. We report this as evaluator path-sampling variance, not as a model change.

3. Implementation and Reproducibility

Table 5. Software, hardware, and external model resources.

Component	Specification
Language and framework	Python 3.10; PyTorch 2.3; CUDA 11.8
Supporting libraries	torchvision 0.18; NumPy 1.26; SciPy; scikit-image 0.23; Pillow 10
Hardware	1 NVIDIA RTX 3060, 12 GB; Intel i7-11700K; 16 GB RAM; Linux
HRVRL weight	G_pretrain.pkl SHA-256: 115eea6d33471910610a13685ae02909 4d1da4e497071c5aad95a49a05acf7e6
RRWNet weight	rrwnet_RITE_refinement.pth SHA-256: 9881bf5c86c94da9099fc141fff3f738 df241c6567f9d366469a83dee54a048f
Frozen output	artifacts/frozen_task1_8_16666/Task1, 50 RGB PNG files
Output manifest	SHA-256: d239841cd10ac62030f1811ec41962dc ff81b531d1c783107f8e4ad88a3ecfa1

The exact Task 1 prediction component is frozen with per-file checksums. All 50 images are 1024 by 1536 RGB PNG files, are zero outside the ROI, and satisfy $\text{vessel} = \max(\text{artery}, \text{vein})$. The five fine-tuned checkpoint hashes, fold manifest, training logs, and source-code hashes are included in the verification package.

Code repository: [insert reviewer-accessible GitHub URL and immutable tag]. Docker image: [insert image, tag, and digest if provided]. Inference entry point: [insert final one-command invocation].

4. Compliance and Limitations

Task 1 is CFP-only. Public HRVRL and RRWNet pretrained weights are disclosed by repository, filename, licence, and checksum. Their use must match the organizer's external-resource policy. The uploaded multi-task archive was later overwritten during submission assembly, so this report does not invent a full archive checksum. The exact Task 1 members remain preserved and individually hashed.

5. References

- [1] GAVE2 Challenge organizers. GAVE2 Challenge, MICCAI 2026 OMIA Workshop.
- [2] HRVRL authors. HRVRL: An Efficient and Interpretable Foundation Model for Retinal Image Analysis in Disease Diagnosis. <https://github.com/sulab-wmu/HRVRL>.
- [3] Morano J, Aresta G, Bogunovic H. RRWNet: Recursive Refinement Network for effective retinal artery/vein segmentation and classification. *Expert Systems with Applications*. 2024;256:124970.
- [4] Liu Z, Mao H, Wu C-Y, Feichtenhofer C, Darrell T, Xie S. A ConvNet for the 2020s. *Proceedings of CVPR*. 2022.
- [5] Hu Q, Abramoff MD, Garvin MK. Automated separation of binary overlapping trees in low-contrast colour retinal images. *MICCAI*. 2013.